

Liang-Ching Tao

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Research Interests

AI planning and sequential decision-making; reinforcement learning; learning-guided search and neural-guided Monte Carlo Tree Search; graph-based policy representations.

Education

National Chung Hsing University (NCHU), Taiwan — M.S. in Computer Science and Engineering Aug. 2026

Advisor: Prof. Pi-Chung Wang. Graduate GPA: 3.84/4.00. Thesis: *A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks*.

National Chengchi University (NCCU), Taiwan — B.S. in Computer Science 2023

Papers and Presentations

- L.-C. Tao and P.-C. Wang, “A Hierarchical Q-learning and Force-aware Routing Approach for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks,” *2026 MCBWASN Conference, Department of Computer Science and Engineering, National Chung Hsing University, Taiwan, Aug. 24–25, 2026. Poster presentation*.
- **Liang-Ching Tao**, Wei-Cheng Tai, Pi-Chung Wang. “Adaptive Charging Threshold and Detour Scheduling for Wireless Rechargeable Sensor Networks via Deep Reinforcement Learning.” *Accepted for oral presentation, MAMM 2026, Sun Moon Lake Teachers’ Hostel, Taiwan, Nov. 5–7, 2026*.
- **Liang-Ching Tao**, Pi-Chung Wang. “HQARRF: Hierarchical Q-learning and Force-aware Routing for Multi-Charger Scheduling in Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation, MAMM 2026, Sun Moon Lake Teachers’ Hostel, Taiwan, Nov. 5–7, 2026. [arXiv PDF]*
- **Liang-Ching Tao**, Po-Cheng Fu, Pi-Chung Wang. “DTSC: A Dual-Tier Sector-Based Cooperative Mobile Charging Scheme for Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation, MAMM 2026, Sun Moon Lake Teachers’ Hostel, Taiwan, Nov. 5–7, 2026*.
- **Liang-Ching Tao**, Wei-Chen Lin, Pi-Chung Wang. “CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks.” *Accepted for poster presentation, MAMM 2026, Sun Moon Lake Teachers’ Hostel, Taiwan, Nov. 5–7, 2026*.

Research Experience

Learning-Guided Planning in Large Dynamic Action Spaces: Budgeted Tree Search for One-to-Many Mobile Charging 2026–present

- Develop LP-BTS, combining a graph proposal policy and learned value critic with edge-budgeted PUCT for one-to-many mobile charging.
- Study sampled candidate support in a large, dynamic, geometrically structured action space; evaluate it with matched ablations and a fixed, pre-specified set of 30 held-out evaluation scenarios. [arXiv PDF]

Learning-Guided Multi-Charger Planning with Neural-Guided Search Sep. 2026–present

- Extend LP-BTS toward multi-charger scheduling with graph-based policy representations, canonical actions, action-space reduction, and neural-guided search.

Hierarchical RL for Multi-Charger Scheduling — M.S. thesis, NCHU 2025–2026

- Formulated coupled charging, routing, and allocation as a structured sequential decision problem in WRSNs; developed a two-level architecture in which Q-learning selects regions or scheduling decisions and force-aware routing determines charger movement.

Other Research Experience

Research Assistant — Raman Spectral Analysis and Detection System, NCHU 2025–2026

- Developed a 1D residual-convolutional and attention-based workflow for Sudan Red I–IV detection, with FastAPI/Next.js integration. [code]

Research Assistant — Citrus Crop Disease and Pest Recognition, NCHU 2025–2026

- Trained and compared EfficientNetV2-S and ResNet50 classifiers; used Focal Loss, MixUp/CutMix, Grad-CAM, and ONNX export for integration. [code]

Teaching Experience

Teaching Assistant, NCHU 2025

Computer Networks (undergraduate and professional master’s program); New Trends in Network Technology (graduate). Instructor: Prof. Pi-Chung Wang. Grading, student Q&A, laboratory assistance, and exam proctoring.

Technical Skills

Python; PyTorch; ONNX; FastAPI; Next.js.